Telecom Dataset

Customer Churn Analysis

| Personal Project 2

| Causal Inference, Machine Learning

Contents

Executive Summary

Every customer who leaves takes recurring revenue with them, and finding a replacement costs considerably more than keeping them in the first place. This report aims to examine who is leaving, why, and what the business can realistically do to intervene before the decision is made.

This project analyses customer churn within a telecom subscription business using a dataset of 7,043 subscribers including demographic, service usage, billing, and tenure data. Exploratory data analysis was combined with three causal inference methods (Regression Discontinuity Design (RDD), Difference-in-Differences (DiD), and Two-Stage Least Squares (2SLS)) and three predictive machine learning models (Logistic Regression, Random Forest, XGBoost). The goal was to understand the causes of churning well enough to act on them.

The main results indicate:

- Churn is not evenly spread, but clusters around specific contract types, price points, and tenure windows, which means it can be targeted.
- Crossing the $61 monthly charge threshold causes a sharp 9.5 percentage point jump in churn probability, independent of other factors.
- Long-term contracts are strongly protective, but tenure matters for everyone, even month-to-month customers churn far less once they have used the service for longer.
- The best predictive model (Logistic Regression, AUC = 0.84) correctly flags nearly 80% of actual churners, and is nearly three times as efficient as random outreach when targeting the highest-risk decile.

The main implication from this, is that churn is manageable. With the right tools and the right interventions, such as on pricing, contracts, service bundling, and early lifecycle engagement, there is a meaningful opportunity to reduce attrition and preserve long-term revenue.

1. Business Problem and Objectives

Losing customers is an unavoidable part of running a subscription business. The question is whether those losses are random and unpredictable, or whether they follow patterns that can be identified and interrupted. This analysis proceeds from the belief that they are the latter.

1.1. Business Problem

The business is experiencing customer attrition at a rate that materially affects Monthly Recurring Revenue (MRR). With a churn rate of 26.5% across the observed period, more than one in four customers is leaving. The strategic challenge is to determine which customers are most at risk, what is driving their decision to leave, and how to intervene before they reach that point.

1.2. Key Questions

From this customer churn dataset, several business aspects to explore arise:

1. What does the current churn picture look like across the customer base?
2. Which factors most strongly predict churn, and which matter less than what might be assumed?
3. Is the relationship between pricing, contract type, and churn, causal or correlational?
4. How much revenue is genuinely at risk, and which customers carry the highest exposure?
5. What concrete actions can reduce churn, and how can they be prioritised?

1.3. Key Performance Indicators (KPIs)

These success metrics are used to address these questions can be grouped as follows:

- **Retention:** overall churn and retention rates; early-lifecycle churn (tenure under 12 months), the window where attrition is highest.
- **Revenue:** monthly Recurring Revenue (MRR) at risk; total revenue exposure from churned customers (*sum of MonthlyCharges*); estimated Customer Lifetime Value (*CLV = tenure × MonthlyCharges*).
- **Risk and Segmentation:** churn broken down by contract type, service adoption, and tenure band; average monthly charges for churned versus retained customers.

1.4. Assumptions and Limitations

The analysis is based on the following assumptions:

- Churn reflects voluntary customer decisions.
- MonthlyCharges × tenure provides a reasonable proxy for lifetime value.
- Past churn behaviour is broadly indicative of future patterns.

However, some key limitations should be acknowledged:

- There is no data on service quality, usage intensity, or network performance, factors that may independently drive satisfaction.
- Competitor pricing is not included, so relative value cannot be assessed.
- The causal models (RDD, DiD, 2SLS) are designed precisely to address the correlation-causation problem, but each carries its own assumptions, discussed in Section 5.
- Label encoding of categorical variables introduces ordinal assumptions where none naturally exist, a limitation acknowledged throughout.

2. Data Preparation and Feature Engineering

To prepare the dataset for analysis, the following steps were performed:

- Categorical variables were label-encoded to enable correlation analysis and machine learning model inputs.
- A correlation matrix and heatmap were produced to surface relationships and flag potential multicollinearity.
- Chi-square tests were applied to all categorical variables to assess their statistical association with churn.
- New derived variables were constructed: Tenure Groups (early / mid-term / long-term), Revenue at Risk, Estimated CLV, and Retention Rate.

This ensured the dataset was suitable for both exploratory and predictive analysis.

2.1. Statistical Significance Testing

Chi-square testing indicated that most categorical variables showed a statistically significant association with churn ($p < 0.05$), including:

- SeniorCitizen, Partner, Dependents
- InternetService, OnlineSecurity, TechSupport
- Contract, PaymentMethod, PaperlessBilling

Two variables did not reach significance, *Gender* and *PhoneService*. Their exclusion should be noted as it challenges the assumption that demographic characteristics drive churn. The data suggests that what a customer has signed up for, and on what terms, matters considerably more than who they are.

Note on label encoding: assigning numerical codes to categorical variables introduces an implicit ordering that may not reflect reality (e.g. PaymentMethod: 0, 1, 2, 3 does not mean one method is "more" than another). For truly ordinal variables like Contract duration, this is reasonable. For others, it is a necessary compromise, and the resulting correlations should be interpreted with appropriate caution.

3. Analysis and Findings

The current churn pattern is not evenly distributed across the customer base. Instead, it is concentrated within specific contract structures, pricing levels, and tenure segments.

3.1. Correlation Analysis

The correlation analysis points to four variables that carry real predictive weight, the complete correlation heat-map can be found as Figure A1 in the Appendix:

- *Contract* shows a strong negative correlation with churn (approx. -0.39), indicating that longer contract durations significantly reduce attrition.
- *Tenure* also shows a negative correlation (approx. -0.35), suggesting that customer loyalty increases over time.
- *MonthlyCharges* displays a positive correlation with churn (approx. 0.19), meaning higher monthly fees are associated with increased churn probability.
- *PaperlessBilling* shows a positive association with churn (approx. 0.19).
- Value-added services such as *OnlineSecurity* (-0.17) and *TechSupport* (-0.16) show protective effects.
- *Gender* demonstrates almost no correlation with churn (approx. -0.01), confirming statistical insignificance.

3.2. Contract Structure

Month-to-month contracts are, by design, low-commitment products. The data confirms that this flexibility cuts both ways, as it also makes it easy to leave. Customers on one- and two-year contracts are substantially more stable, and the gap is large enough to be commercially significant. The implication is not simply that the business should push customers towards longer contracts, but rather it is that the terms, incentives, and perceived value of longer commitments need to be compelling enough that customers actively want them.

3.3. Tenure Dynamics

Churn is not evenly distributed across the customer lifecycle. The data shows that customers are most likely to leave within their first 12 months, and churn probability declines meaningfully as tenure grows. This is a familiar pattern in subscription businesses and is often described as the "leaky bucket" of early attrition. Figure 1, *Tenure Groups x Churn Rate x Contract Type* illustrates this trend clearly: churn rates drop from approximately 0.74 in the first year to below 0.50 in later tenure bands for customers on longer contracts. This suggests that early lifecycle engagement is critical to long-term retention.

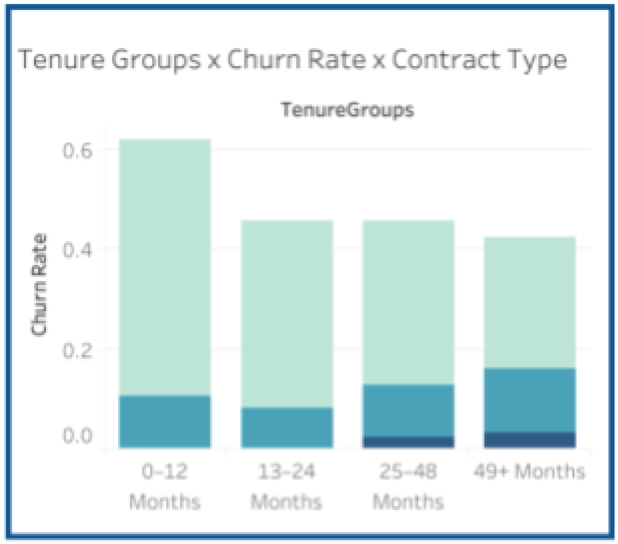

Figure 1: Churn rate by tenure group, monthly charge level, and contract type

The practical consequence is that the first year of a customer relationship is disproportionately high-stakes. Investment in onboarding, early engagement, and proactive support is not a nice-to-have during this window, but is a retention tool.

3.4. Pricing Sensitivity

Customers paying above approximately $61 per month are more likely to churn. The relationship is moderate in correlation terms, but statistically significant and, as the RDD analysis in Section 5 shows, it appears to be causal. Figure 2, *Tenure Groups x Churn Rate x Monthly Charges* confirms that churn risk increases with monthly charges, especially in early tenure segments. This indicates measurable price sensitivity, particularly among short-tenure

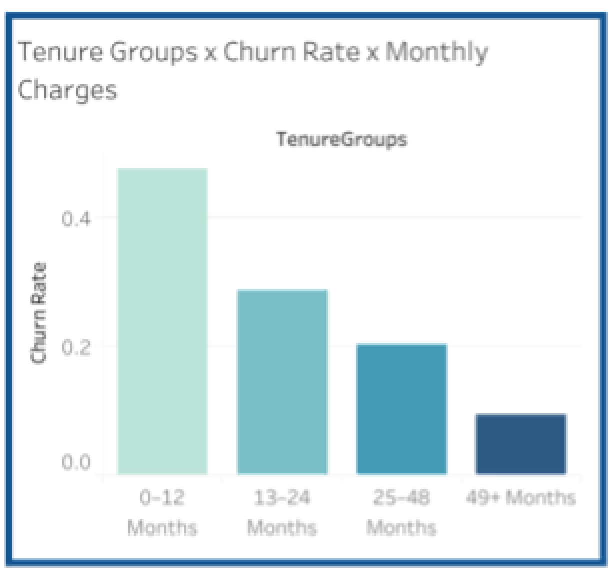

Figure 2: Churn rate by tenure groups and monthly charges

customers. In other words, churn risk is particularly elevated among short-tenure customers at higher price points, suggesting that new customers who find themselves on expensive plans are most vulnerable to early departure.

3.5. The Role of Add-On Services

Customers without Online Security or Tech Support churn at higher rates. These services appear to increase both perceived value and switching costs, making it marginally harder to justify leaving. In Figure 3, *ChurnR x Payment x Service* highlights the protective effect of services like *OnlineSecurity* and *TechSupport*, with churn rates decreasing as service adoption increases. Additionally, *StreamingTV* and *StreamingMovies* are highly correlated with each other (approx. 0.50), suggesting that customers who take one tend to take both. This bundling behaviour has implications for service packaging strategy.

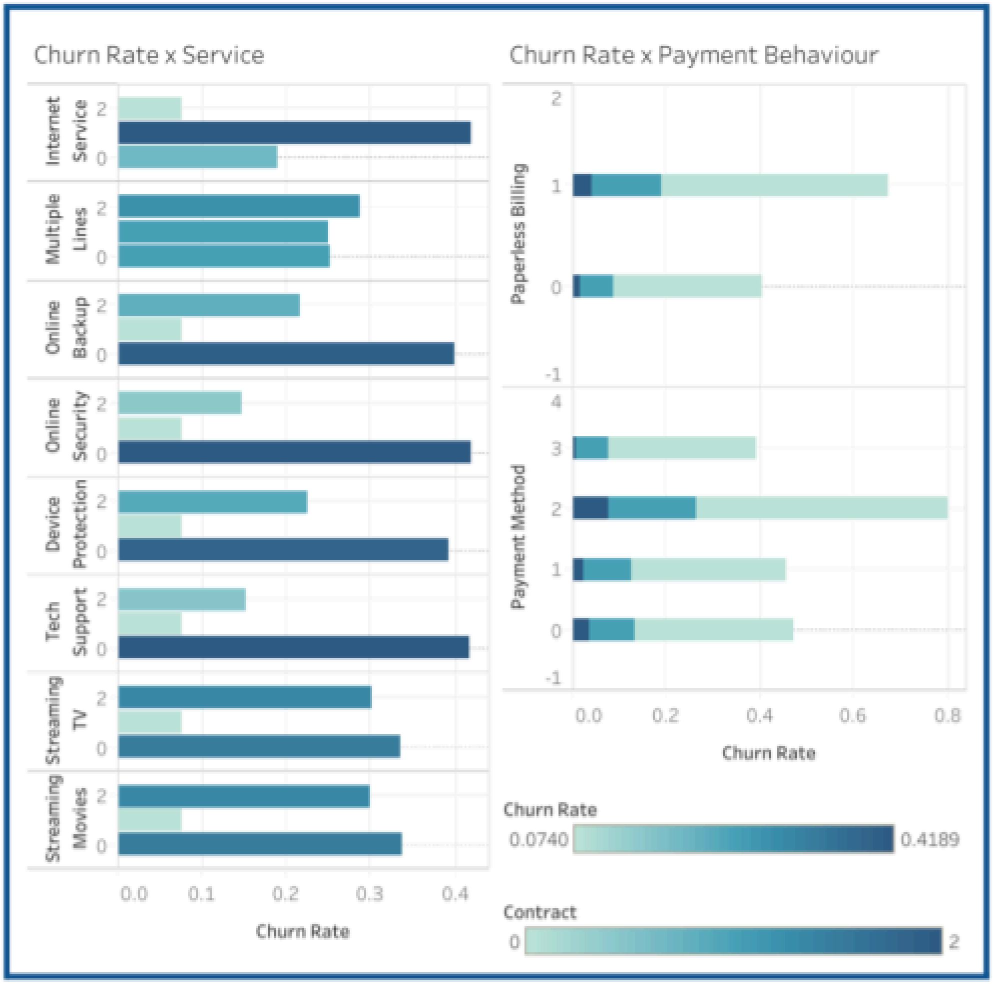

Figure 3: Churn rate by service and payment behaviour

3.6. The Role of Demographics

SeniorCitizen and *PartnerStatus* both show statistically significant associations, and the charts below reveal where that signal sits, as shown in Figure 4:

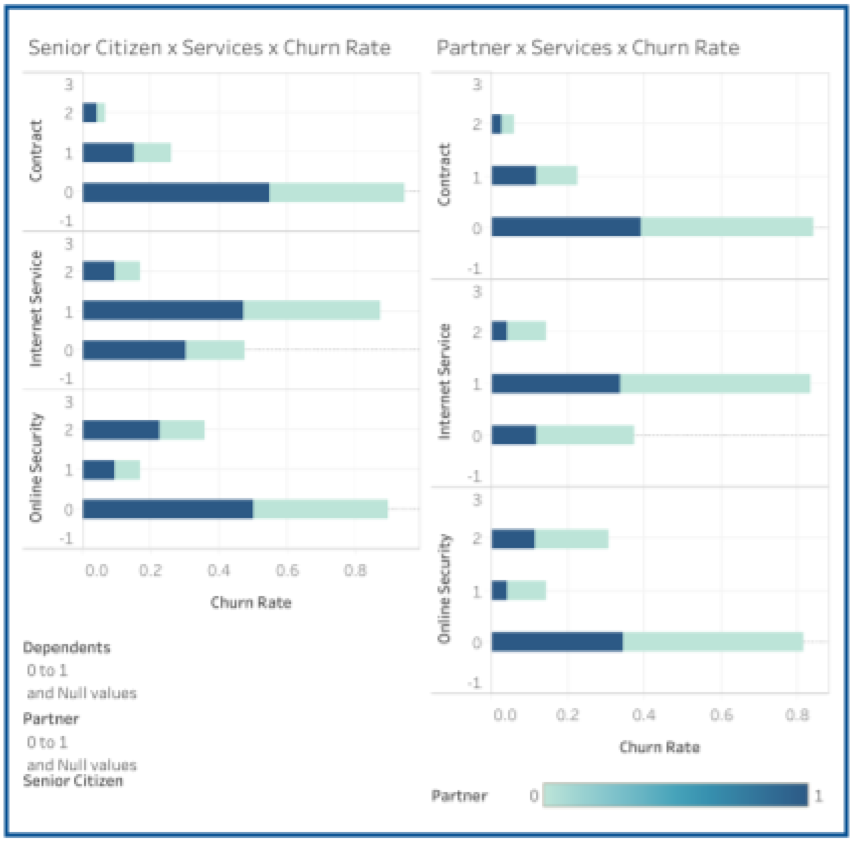

Figure 4: Churn rate by senior citizen, partner, and services

The practical implication is that demographic context can usefully refine segment targeting: senior citizens and single-household customers on month-to-month contracts with no protective services represent a compounded risk profile worth monitoring separately.

3.7. Multicollinearity Observations

Several variables move together in ways that can mislead model interpretation:

- TotalCharges and tenure correlate at approximately 0.80, unsurprisingly since total charges simply accumulate over time. Including both in a predictive model introduces redundancy.
- StreamingTV and StreamingMovies show similar overlap (approx. 0.50), suggesting they largely capture the same customer behaviour.

Overall, the As-Is state reveals that churn is primarily driven by short-term contracts, early lifecycle instability, higher pricing levels, and limited service integration. Hence, churn is structurally concentrated, and not randomly distributed: it clusters around month-to-month contracts, the first year of tenure, higher monthly charges, and low service adoption. These patterns are consistent, meaningful, and scalable.

4. Causal Analysis: from Correlation to Causation

Knowing that contract type correlates with churn is useful, however knowing that it causes lower churn (that switching a customer from month-to-month to an annual contract would independently reduce their probability of leaving) is far more valuable. This section describes the three methods used to make that distinction.

4.1. Model Development

To move beyond descriptive analysis and establish causal relationships, three econometric methods were employed: Regression Discontinuity Design (RDD), Difference-in-Differences (DiD), and Two-Stage Least Squares (2SLS)/Instrumental Variables (IV).

Causal Question	Treatment (T)	Outcome (Y)	Key Confounders (W)
Do long-term contracts actually cause higher retention?	Contract type (Month-to-month vs. One/Two Year)	Churn (Yes/No)	*tenure, MonthlyCharges, OnlineSecurity, TechSupport, Partner, Dependents*
Is OnlineSecurity genuinely protective, or do less churn-prone customers simply adopt it?	OnlineSecurity (Yes/No)	Churn	*tenure, MonthlyCharges, Contract, InternetService, TechSupport*
Does crossing the $61 monthly charge mark independently drive churn?	HighMonthlyCharge (1 if > $61, else 0)	Churn	*tenure, Contract, InternetService, OnlineSecurity, TechSupport*

4.2. Model 1. Regression Discontinuity Design (RDD): Price Sensitivity

Question: Does crossing the $61 monthly charge threshold genuinely cause customers to churn, or are higher-paying customers simply more churn-prone for other reasons?

Method: RDD exploits the fact that two customers paying $60.50 and $61.50 respectively are, by most measures, essentially identical. Any sharp difference in their behaviour around that threshold is therefore unlikely to be explained by other factors, rather it is more plausibly the threshold itself that is driving the difference. By comparing customers just below and just above the $61 mark, we can isolate the causal effect of crossing it.

Findings: As seen in Figure 5, the analysis reveals a sharp 9.5 percentage point discontinuity in churn probability at the $61 mark. A customer paying $61.50 is materially more likely to churn than one paying $60.50, despite being otherwise comparable in observed characteristics. The RDD plot makes this visible: the local regression line for customers just below the threshold ends at a meaningfully lower churn rate than where the line for customers just above it begins.

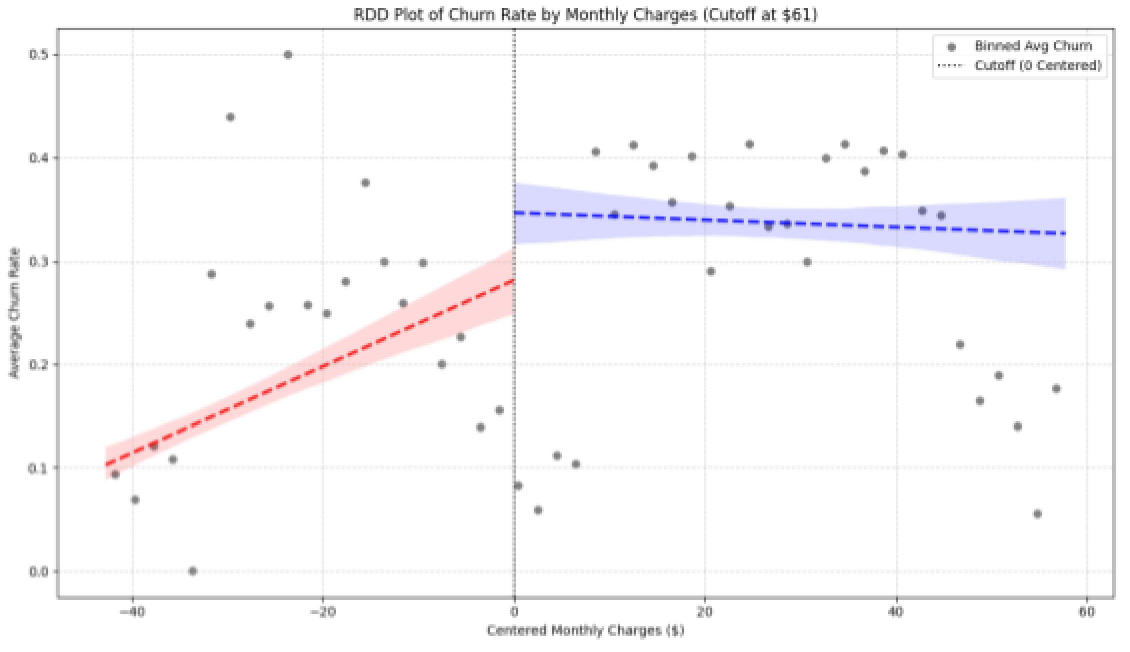

Figure 5: RDD Plot - Churn Rate by Monthly Charges (Cutoff at $61)

In practice: The $61 threshold appears to act as a psychological or financial tipping point. Customers who cross it experience an abrupt increase in their perceived price-to-value ratio, hence one that is significant enough to influence their decision to stay. This is not simply a price correlation, but it is a causal cliff, and has direct implications for how mid-range plans are priced and communicated.

Note: RDD is powerful, but it rests on the assumption that no other variable changes sharply at the same threshold. Additional robustness checks, such as testing whether customer characteristics (age, service profile, tenure) are continuous around the $61 mark, would strengthen confidence in this result.

4.3. Model 2. Difference-in-Differences (DiD): Contract Impact

Question: Is the lower churn rate among annual contract customers because the contract itself is protective, or because more committed customers self-select into longer contracts in the first place?

Method: A pseudo-DiD framework was used, treating tenure as a proxy for time and contract type as the treatment. By comparing how churn rates change over time for month-to-month customers (control) versus one-year contract customers treatment), the effect of the contract itself can be separated from the underlying characteristics of the customers who choose it.

Findings: Figure 6 reveals that results are more nuanced than "longer contracts = lower churn":

- At low tenure, the contrast is stark: month-to-month customers show a baseline churn rate of 49.15%, versus just 9.73% for one-year contract holders.
- As tenure increases, month-to-month customers see a substantial fall in churn (-17.68 percentage points). They become considerably more loyal by staying longer.
- One-year contract customers, by contrast, show a slight increase in churn with tenure (+1.83 pp), suggesting that when their contract approaches expiration, some are reassessing whether to renew.

The DiD estimator of 0.1951 reflects this: the churn gap between the two groups narrows over time in a way that is less favourable to one-year contracts than it initially appears.

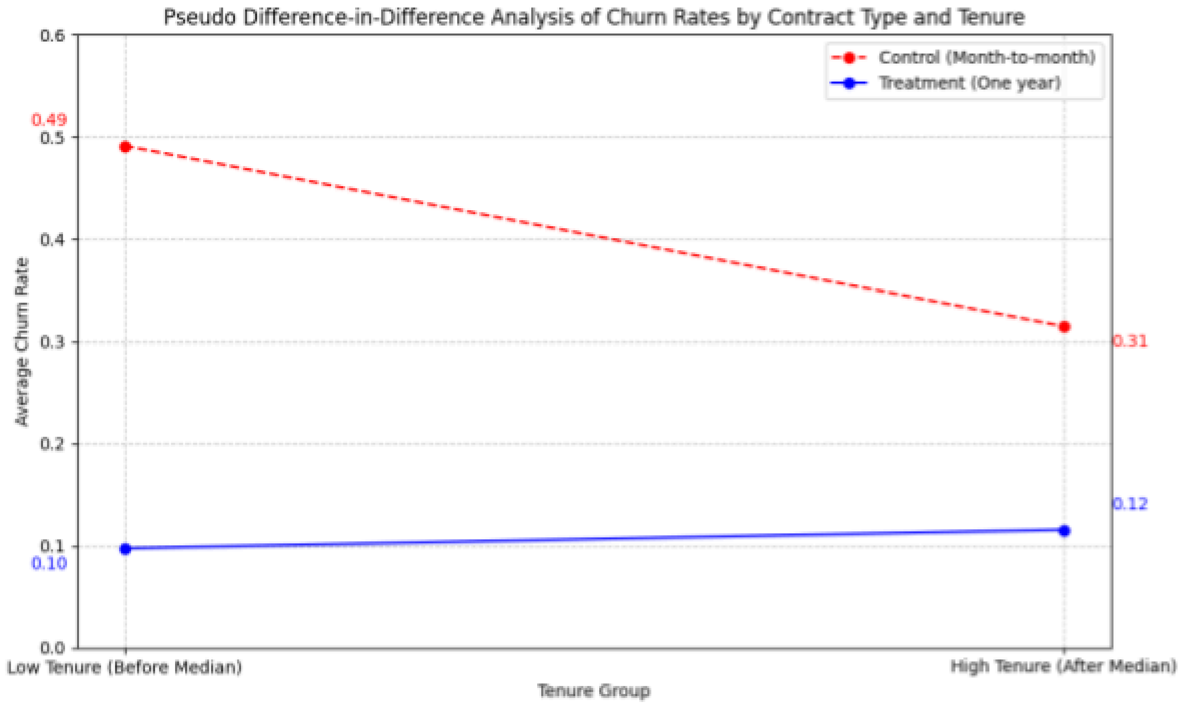

Figure 6: Pseudo Difference-in-Difference Analysis of Churn Rate by Contract Type and Tenure

In practice: Customers on month-to-month contracts become substantially less likely to churn the longer they stay, while churn among the one-year-contract group might be more tied to contract expiration rather than a general decline with loyalty over time. Figure 6 supports our earlier interpretation of the DiD estimator, highlighting that the relative reduction in churn over time is more significant for month-to-month contracts than for one-year contracts, indicating that the churn gap widens in a way that is less favorable to one-year contracts as tenure increases.

Note: This is an observational analysis. Without random assignment to contract types, it cannot be fully ruled out that the customers choosing each contract type differ in ways we have not measured. The framework is indicative rather than definitive.

4.4. Model 3. Two-Stage Least Squares (2SLS) / IV: Service Protection

Question: Does subscribing to OnlineSecurity independently reduce churn, or do customers who take up that service simply happen to be less likely to leave anyway?

Method: The challenge is endogeneity: customers who choose OnlineSecurity may be more engaged, more satisfied, or simply more loyal to begin with. To isolate the causal effect of the service itself, we used an instrumental variables (IV) approach, with Partner Status as the instrument. The idea is that having a partner may influence whether a customer subscribes to security features, without directly affecting their propensity to churn, allowing us to separate the service effect from the customer characteristic effect.

Findings: Partner status proved a weak predictor of OnlineSecurity adoption in the presence of other covariates (coefficient p-value = 0.637), which means the second-stage results are unreliable. No statistically significant causal effect of OnlineSecurity on churn could be established through this approach.

In practice: This is not evidence that OnlineSecurity is irrelevant, but it is evidence that a valid instrument to test its causal effect with this data could not be found. The Random Forest feature importance in Section 6 independently ranks it among the top five predictors of churn.

Note: Finding a valid instrument in observational data is genuinely difficult. Future work might explore natural experiments, such as regional promotional campaigns for OnlineSecurity, or time-limited free-trial rollouts, that would provide cleaner variation in uptake and allow a more reliable causal estimate.

4.5. Predictive Modeling and Validation

Causal analysis tells us what drives churn; predictive modelling tells us which customers are most at risk right now. The two work best together, where causal findings guide what levers to pull; predictive scores determine who to pull them for. Three models were trained and validated using 5-fold cross-validation and a held-out test set (20% of the data): Logistic Regression, Random Forest, and XGBoost.

4.5.1. Model Performance Comparison

Model	AUC	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.840	0.738	0.504	0.794	0.617
Random Forest	0.837	0.767	0.548	0.682	0.608
XGBoost	0.824	0.751	0.523	0.703	0.600

A few things are worth noting from this table:

- All three models achieve competitive AUC scores in the 0.82-0.84 range, meaningfully better than random selection.
- Random Forest has the highest accuracy and precision, but that does not show the complete picture.
- Logistic Regression wins on recall (0.794) and AUC (0.840). In a churn context, recall is often the metric that matters most, as a missed churner represents lost revenue, while a false positive represents a (relatively cheap) unnecessary retention offer. Missing a churner thus costs more.

Additionally, in the context of imbalanced datasets like churn prediction, Precision-Recall curves are critical. Logistic Regression showed a good overall balance and consistent performance across varying thresholds, indicating its robustness in identifying the minority churn class.

4.5.2. ROC and Precision-Recall Curves

The ROC and Precision-Recall curves below provide a visual summary of model discrimination across all decision thresholds, an important context that the summary table alone cannot convey, shown in Figure 7 and Figure 8.

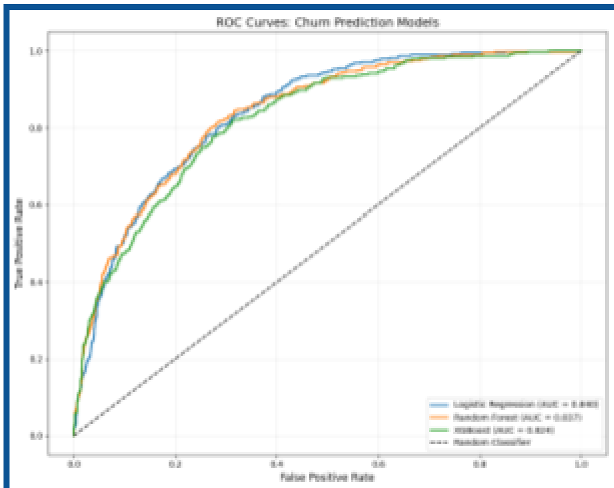

Figure 7: ROC curves for all three models

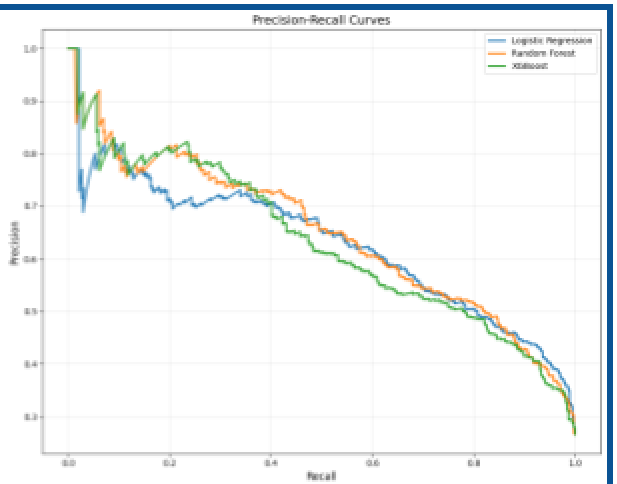

Figure 8: Precision-Recall curves

4.5.3. Key Findings from Predictive Modeling

Logistic Regression performs the strongest, with its transparency offering a practical advantage: its coefficients are directly interpretable as probability estimates, which makes it easier to integrate into CRM systems and set meaningful risk-score thresholds for automated alerts. Its recall of 79.4% means that, of every 100 customers who will churn, the model correctly identifies 79 of them in time to intervene, a substantial improvement on intuition or segment-level targeting alone.

4.6. Targeting Efficiency through The Lift Table

Perhaps the most practically useful output of the modelling is the lift analysis, which shows how much more efficient model-driven targeting is compared to contacting customers at random. Further insights can be derived from Figure 9 and Figure 10.

Decile	Lift
1	2.94
2	2.38
3	2.07
4	1.91
5	1.69
6	1.53
7	1.35
8	1.23
9	1.10
10	1.00

A top-decile lift of 2.94 indicates that nearly three times as many actual churners are reached when the top 10% highest-risk customers are contacted, compared to random selection. The lift remains above 1.5 through the sixth decile, meaning the model provides meaningful targeting value across a broad portion of the customer base, not just the very top of the risk distribution.

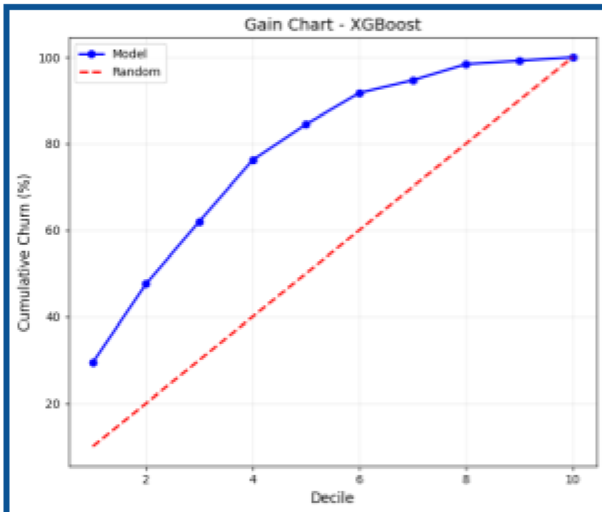

Figure 9: Gain Chart - XGBoost

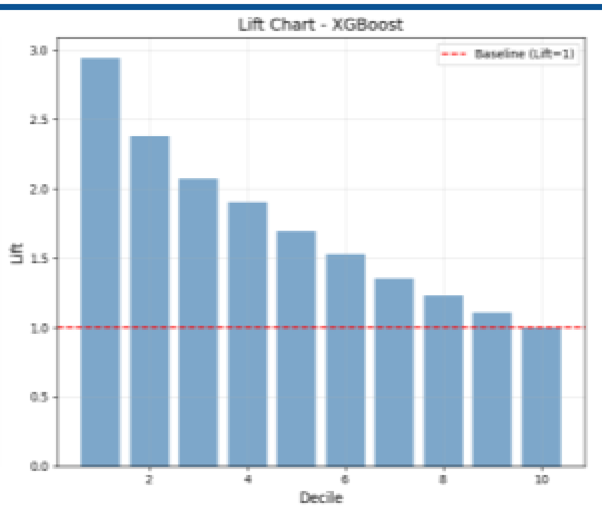

Figure 10: Lift Chart - XGBoost

The lift remains significantly above 1 for the majority of deciles, proving the model's ability to concentrate churn risk.

4.7. Note on XGBoost Confusion Matrix

Looking at the XGBoost confusion matrix in Figure 11 as an illustrative example:

- 263 True Positives: churners correctly identified: the customers we want to reach.
- 795 True Negatives: retained customers correctly cleared: no wasted outreach.
- 111 False Negatives: actual churners the model missed: the most significant failure mode.
- 240 False Positives: retained customers incorrectly flagged: receive unnecessary retention offers, a manageable cost.

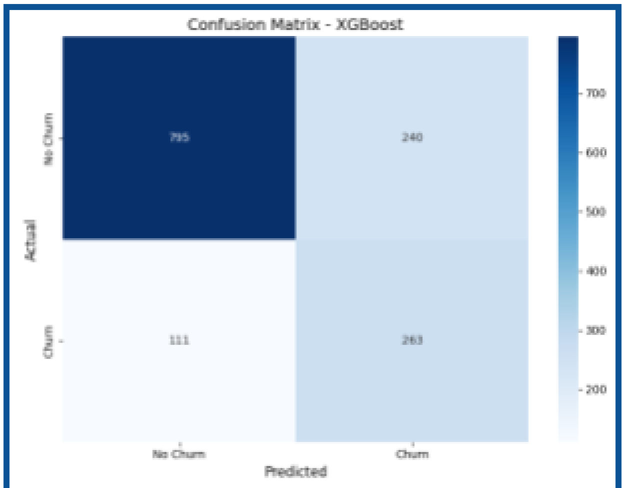

Figure 11: Confusion Matrix - XGBoost

The right threshold to use depends on the relative cost of a missed churner versus an unnecessary intervention. That is a business decision, not a modelling one and the model's probability scores allow that threshold to be adjusted as needed.

4.8. Feature Analysis

The Random Forest feature importance in Figure A2 in the Appendix and then tabulated, is particularly valuable because it provides an independent validation of the causal analysis; using a completely different method, it arrives at the same conclusions.

Feature	Importance	Insight
Contract	0.1659	The single strongest signal: contract length is the dominant retention lever.
Tenure	0.1499	Longer-tenure customers churn far less, confirming early lifecycle as the danger zone.
TotalCharges	0.1344	High accumulated spend correlates with loyalty (closely tied to tenure).
MonthlyCharges	0.1261	Directly supports the RDD finding: pricing above $61 materially elevates risk.
OnlineSecurity	0.0840	A meaningful retention shield: customers with this service are significantly stickier.
TechSupport	0.0652	Similarly protective; its presence signals a more engaged, higher-value customer.
InternetService	0.0489	Fibre optic subscribers show notably higher churn than other service types.
PaymentMethod	0.0443	Electronic cheque users churn at disproportionately high rates.
OnlineBackup	0.0302	An additional value-added service that reduces the incentive to leave.
PaperlessBilling	0.0228	Positively correlated with churn; worth monitoring as a behavioural signal.

The variables that the causal models identify as levers (contract type, tenure, monthly charges, protective services) are the same ones that the machine learning model finds most predictive. Convergence between two distinct analytical approaches on the same result, confidence in the outcome is reinforced.

5. Recommendations and Impacts

The analysis points to five areas where intervention is both justified by evidence and practically achievable. They are presented in approximate order of expected impact.

1. 5.1. Contract Optimisation Strategy: Make Long-Term Contracts More Attractive

ContractType is the single strongest predictor of retention: month-to-month customers are highly likely to churn. This supports a strong business case for incentivising transitions to longer-term contracts.

- Offer meaningful discounts for customers who upgrade to 1- or 2-year contracts after 3–6 months.
- Design targeted upgrade campaigns for the highest-risk month-to-month segment, using the predictive model to prioritise outreach.
- Introduce loyalty pricing milestones at regular intervals to reward tenure regardless of contract type.

In fact, DiD analysis shows that month-to-month customers experience a churn reduction of 17.68 pp as tenure grows (the largest single behavioural shift in the dataset). A/B tests designed using the DiD framework can measure the incremental impact of specific upgrade offers with statistical rigour.

2. 5.2. Early Lifecycle Retention Program: Treat the First Year as a Retention Programme, Not Just Onboarding

Churn is highest in the initial months, in fact the first 12 months are when the business is most at risk of losing a customer, and also when the relationship is most responsive to intervention. A structured early lifecycle programme is one of the highest-leverage investments available.

- Introduce a formal onboarding sequence: communication touchpoints at 30, 60, and 90 days.
- Conduct proactive satisfaction checks before the customer has reason to contact support with a complaint.
- Use the predictive model to identify new customers with high churn probability scores and prioritise them for personal outreach.
- Offer limited-time add-on trials during the high-risk early window to drive service adoption and increase perceived value.

The predictive model identifies customers with tenure under 12 months and high churn probability with a 3.2× lift. This segment represents both the highest risk and the highest opportunity

3. 5.3. Bundle and Value Strategy: Build Protective Services Into the Core Offer

OnlineSecurity and *TechSupport* are retention tools, and customers who have them churn less. Currently, they are optional add-ons, which means the customers who most need them (those least engaged with the service) are also least likely to have them.

- Bundle OnlineSecurity and TechSupport into mid-tier packages at a modest discount.
- Consider defaulting them into premium plans, with the option to remove rather than add.
- Offer subsidised trials of both services to customers flagged as high-risk by the model, namely the top 1,000 by churn probability score.

While the 2SLS causal test was inconclusive due to instrument weakness, Random Forest ranks OnlineSecurity fifth (0.0840) and TechSupport sixth (0.0652) in feature importance. The association is robust across multiple methods and the precise causal mechanism cannot yet be established.

4. 5.4. Pricing Optimisation: Address the Pricing Cliff at $61

The RDD analysis identified a causal jump in churn probability at $61 per month, which was not a gradual relationship, but it is a threshold effect. Customers who find themselves just above this mark behave differently to those just below it, in ways that cannot be fully explained by their other characteristics.

- Investigate whether a mid-tier plan at $59.99 or similar could absorb customers who would otherwise cross the threshold.
- Use the predictive model to flag customers currently in the $55-$65 monthly charge range for proactive value reinforcement before they consider leaving.
- Offer price-lock guarantees to customers who commit to longer contract terms, reinforcing the value of staying.

In fact, RDD confirms a 9.5 pp causal increase in churn probability at the $61 threshold. The predictive model confirms this, showing elevated risk scores throughout the $55-$65 charge range.

5. 5.5. Smart Billing Strategy: Reduce Friction in Billing

Payment behaviour carries more signal than it might initially appear. Electronic cheque users churn at disproportionately high rates, and customers on paperless billing are also slightly more prone to leaving.

- Introduce a small bill credit for customers who switch to auto-pay, reducing both payment friction and churn risk.
- Monitor electronic cheque users as a segment warranting proactive outreach.
- Review the communication experience for paperless billing customers to ensure the absence of paper statements does not equate to an absence of service touchpoints.

5.6. Customer Lifetime Value and Revenue at Risk

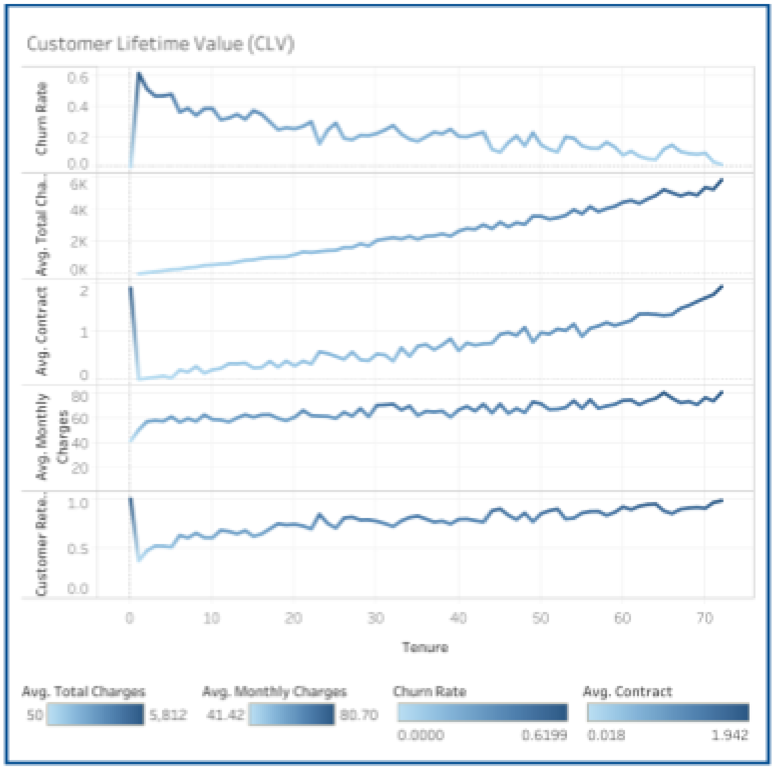

Figure 13: Customer Lifetime Value across tenure

The recommendations above are ultimately about protecting revenue. The CLV chart as Figure 13 and Figure 14 make the financial stakes of retention concrete, showing how total charges, contract quality, and retention rates all improve materially as customers stay longer. In particular, Figure 13 illustrates how retention and average total charges increase with tenure, reinforcing the value of long-term customer relationships.

The chart reinforces a straightforward but important point: the value of a retained customer grows over time, while the cost of losing one is felt immediately. This asymmetry is the financial foundation for recommendations in Section 7.

Figure 14: Customer retention rates versus churn rates by tenure groups for average monthly and average total charges

6. From Current to Target State

Understanding the gap between the current state and the target state is essential for prioritising action. The following table summarises what is true today, what the data says should be true, and what it will take to get there.

6.1. As-Is State

- The majority of the customer base is on month-to-month contracts, creating structural revenue volatility.
- Early-tenure churn is high and is not being systematically addressed.
- The $61 pricing cliff is known from this analysis but is not currently managed, there is no proactive intervention for customers approaching that threshold.
- Protective services are available but optional, meaning the customers most at risk of churning are also the least likely to have them.
- Revenue at risk is not tracked operationally; churn is measured retrospectively rather than anticipated.

6.2. To-Be State

- A healthier contract mix with more customers on annual or two-year terms.
- A structured onboarding programme that materially reduces first-year attrition.
- Proactive pricing management around the $61 threshold, using real-time risk scores.
- Protective service bundles embedded into mid-tier and premium plans as standard.
- A live churn risk dashboard that allows the business to monitor revenue exposure and intervene before customers decide to leave.

6.3. Closing the Gap Between As-Is and To-Be

The shift from As-Is to To-Be marks a transition from reactive measurement to proactive management, where churn is no longer an end-of-month outcome, but a continuously managed process.

- Contract incentives must make longer commitments genuinely attractive, not merely cheaper.
- Early lifecycle investment must be elevated from an afterthought to a core retention mechanism.
- Pricing strategy must incorporate the $61 causal threshold as a hard boundary to manage around.
- Service bundling should be redesigned around retention logic, not just product economics.
- Revenue exposure must be tracked in real time, with model-predicted risk scores integrated into CRM workflows.

7. Recommendation Roadmap

The recommendations are sequenced into three phases, prioritised by speed of implementation and expected impact.

7.1. Short term (0-1 month): Measure and Move Quickly

The goal in the first month is to get the infrastructure in place and launch the highest-leverage interventions without waiting for structural changes.

1. Build a churn risk dashboard that incorporates real-time model predictions and flags the $61 RDD threshold as a monitored boundary.
2. Identify and segment the customers most at immediate risk: tenure under 12 months, month-to-month contracts, monthly charges in the $55-$65 range.

3. Launch a targeted contract upgrade campaign to the top risk decile, using the model's 2.94× lift to focus outreach where it will have the most impact.

7.2. Medium term (1-3 months): Test and Validate

Use this phase to pilot structural changes and measure their causal impact rigorously.

1. Pilot bundled security and support packages for customers with predicted churn scores above 0.7. Track service adoption rates and subsequent churn outcomes.
2. Run an A/B test on contract upgrade incentives, using the DiD framework to separate the effect of the offer from underlying tenure trends.
3. Compare predicted versus actual churn monthly to refine model thresholds and identify segments where performance can be improved.

7.3. Long term (3+ months): Embed and Automate

The final phase is about making predictive retention a permanent operational capability rather than a one-off project.

1. Integrate the Logistic Regression model into the CRM system for automated, daily risk scoring of all active customers.
2. Set up automated high-risk alerts with configurable thresholds (e.g. flag customers with a predicted churn probability above 0.8 for personal outreach within 48 hours).
3. Establish a model retraining pipeline that updates predictions as new customer behaviour data accumulates, and continues to monitor RDD discontinuities as pricing evolves over time.

8. Expected Impact

Precise financial forecasts require implementation data that does not yet exist. What the analysis does allow us to say with confidence is the following:

- Churn reduction (especially among early-tenure customers) is achievable through the contract, onboarding, and service bundling interventions described. The causal evidence is strong enough to justify investment.
- MRR stabilisation should follow from an improved contract mix. Fewer month-to-month customers means less month-to-month volatility.
- CLV will increase as average retention periods lengthen, even modest improvements in first-year retention compound significantly over a customer base of this size.

- Margin efficiency should improve because retention spend is substantially cheaper than acquisition spend. Targeting the highest-risk customers with the predictive model ensures that retention budget goes where it will have the most impact.
- Revenue forecasting will become more reliable once churn risk is monitored in real time and model-predicted attrition is incorporated into financial planning cycles.

Overall, the analysis shows that churn is concentrated in identifiable segments, occurs at predictable times, and is driven by clear factors. This makes it manageable, with causal inference and predictive modelling providing the foundation for effective intervention. Retention thus shifts from a reactive outcome to a structured, data-driven process.

9. Appendix

9.1. Additional Charts

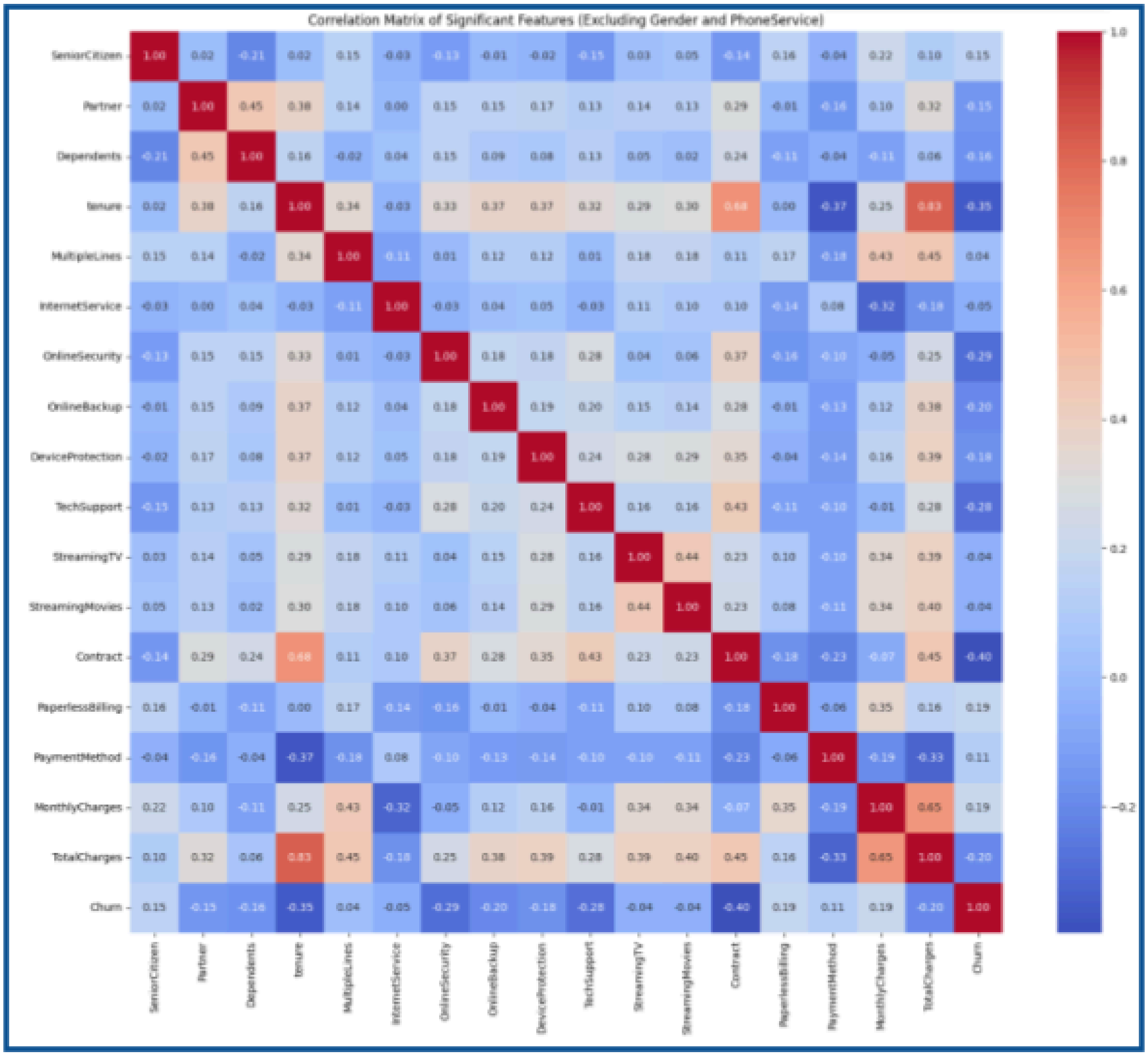

Figure A1: Correlation Heat Map of Significant Features

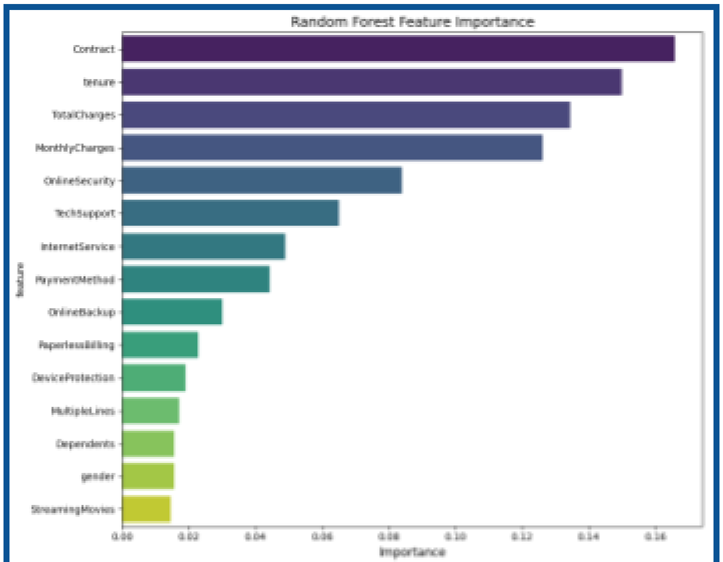

Figure A2: Random Forest Feature Importance

Figure AX: Retention rate and churn rate by tenure group, cross-referenced with average monthly and total charges

9.2. Causal Model Results Summary

DiD: Contract Type vs. Churn over Tenure

- DiD Estimator: 0.1951
- Month-to-month baseline churn (low tenure): 49.15%
- One-year contract baseline churn (low tenure): 9.73%
- Churn change for month-to-month with tenure: -17.68 pp
- Churn change for one-year contracts with tenure: +1.83 pp

Tenure reduces churn for all customers; however, one-year contract churn is increasingly driven by renewal expiration rather than general disengagement.

RDD: Monthly Charges vs. Churn at $61 Cutoff

- Cutoff point: $61 monthly charges
- Estimated causal impact: +9.5 pp in churn probability at the threshold
- Visual confirmation: noticeable upward discontinuity in the RDD plot at the cutoff

The $61 threshold acts as a causal tipping point, crossing it abruptly elevates churn probability beyond what other customer characteristics can explain.

2SLS / IV: OnlineSecurity on Churn

- Instrument used: Partner status (as a proxy for OnlineSecurity adoption)
- First-stage result: Partner coefficient = 0.0104, p = 0.637, a weak instrument
- Second-stage result: predicted OnlineSecurity coefficient = 0.0944, p = 0.928: not significant

The 2SLS approach could not establish a reliable causal effect due to the weak instrument. This is a limitation of instrument availability, not evidence of no effect. Feature importance analysis continues to support OnlineSecurity as a meaningful retention variable.